CLIMATE CARES
Mental health in the climate crisis

TOOLKIT

Climate Change and Mental Health:

An Introductory Toolkit for Researchers

connecting climate minds

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Imperial College London
Projects

An Introductory **Toolkit** for Researchers

Why should we care about the nexus of **climate change** and **mental health**?

“ I think about how physical health is related to mental health and is **intrinsically tied to planetary health** and how we as Indigenous People care for over 80% of the biodiversity on this planet ”

“ We need **advocacy** not only for the health of people but the health for all that exists ”

Shared by one of the Indigenous collaborators in the Connecting Climate Minds Dialogue

Climate change and mental health are two of the greatest global challenges societies face and are deeply interconnected. Climate change impacts mental health by increasing the risk of new mental health challenges and increasing the vulnerability of people with mental health challenges. Individuals and communities increasingly experience climate-related psychological distress and elevated risk of post-traumatic stress disorder, anxiety and mood disorders, substance use problems and suicide. This can be due to compounding stressors from direct experiences of extreme climate and weather events and their aftermath, including food and water insecurity, forced migration, poverty, cultural disruptions and violence. Awareness of current and future hazards associated with the climate crisis and insufficient action from leaders can also cause distress, that while not pathological, can pose risks to mental health and wellbeing.

Climate-induced consequences can hence incapacitate individuals, communities and systems, worsening mental health and well-being and the capacity to cope with and act on climate change in a vicious cycle[1]. The cost of the additional mental health burden from climate-related hazards, air pollution and lack of green space has been estimated at $US 47 billion by 2030, growing to $US 537 billion in 2050[2].

***Figure 1**, adapted from Lawrance et al. 2022[1]: Examples of the continuum of impacts that climate change has on mental health outcomes. Climate change impacts (top row of circles) including rising temperatures and sea level, and extreme weather events such as floods or droughts, effect mental health and emotional wellbeing (bottom row of circles) including: new cases or increased symptoms of mental disorders; increased deaths by suicide; increased susceptibility to physical illness or death for those who meet the criteria for mental disorders; worsened population mental health, and mental and emotional distress. This occurs directly and indirectly via a variety of pathways represented here as a continuum, from direct experiences (left hand side of shaded box), for example of extreme high temperatures or home loss in a wildfire, to indirect experiences of climate impacts (right hand side of shaded box), for example by reading about such events in the media and hearing about insufficient climate action from leaders.*

From a vicious to a virtuous cycle, climate action is an opportunity multiplier for mental health and well-being. Psychological resilience is a key enabler of climate resilience and collective action can protect mental health from climate distress. The conditions we need for climate mitigation and adaptation align with those that support good mental health, such as cleaner air, greener cities, active transport, reduced inequalities, and stronger, more cohesive communities.

There are multiple, interacting pathways by which climate change and mental health interact, with overlapping causes, consequences, and solutions. Now more than ever, research and action on climate change and mental health must bring together diverse expertise across disciplines, sectors, geographies, and lived experiences. Progress to protect the mental health of increasing numbers of people exposed to climate hazards and to enable climate action that brings co-benefits for mental health depends on researchers from diverse backgrounds bringing their perspectives, methods, and datasets. By this, new evidence in the field of climate change and mental health would be developed that can underpin decisions in policy and practice. The nexus of climate change and mental health is also relevant to integrate into work across diverse sectors and disciplines. For example, climate change attribution science could help improve understanding of the relationships between extreme weather and climate events and mental health outcomes. People working in suicide prevention may need to consider how the risk of suicide is affected by rising temperatures and extreme heat, and agricultural researchers may need to consider the mental health impacts of climate change and climate mitigation and adaptation efforts on farming communities.

Examples of climate change impacts on mental health

Increasing temperatures

Extreme heat increases suicide risk and leads to higher emergency room visits by people with mental health challenges[3]. People living with mental health challenges like schizophrenia and depression can be more likely to die in a heatwave, perhaps in part because of psychoactive medication impairing the body's ability to regulate its temperature[4].

Extreme weather and climate events

Acute traumas from extreme weather events and their aftermath can manifest as long-term post-traumatic stress disorder, substance use depression, anxiety, and more deaths by suicide[5]. CCM participants report that the increased frequency of climate hazards is impairing the ability of communities to recover before the next trauma hits.

Destabilisation of mental health determinants

Air pollution, generated by burning fossil fuels and worsened by the climate crisis, has been linked to anxiety, depression, psychosis and increased use of mental health services[6].

Awareness of the climate crisis

Awareness of the climate crisis is causing high distress in children and young people due to betrayal by decision-makers[7], and climate change is experienced as an extension of colonialism by many affected communities[8].

“Mothering a **newborn baby** and **forest fires raging** during this period. I have been in forest fires before but being with my baby in it **gave me so much fear**...”

“My village is in a delta state. Almost every year, **floods come** because the village is in a coastal region and cover the entire village...People who live there and their **livelihoods are affected**. Young people witness all this and have a **sense of helplessness**...”

“One thing we talk about in my spiritual care world is anticipatory creep. You go through the **trauma** of experiencing things, but then the foreknowledge of how bad things can get... You're **already anticipating grieving** even before things actually manifest. It's a part of the heart of people to do so, but it can also really **take a wear and tear** on the spirit and psyche...”

CCM dialogue participants describe the mental health challenges of climate hazards **in their own words**.

Where is research and action needed for the climate change and mental health field?

The impact of climate change on mental health has been noted by a small pool of academics and practitioners for more than a decade. Until recently, research explicitly exploring the 'climate change and mental health' nexus remained relatively scant. A 2021 review of literature explicitly on climate change and mental health found only 120 papers, with the first being published in 2007. Similarly, another review noted that although a search of PubMed for the decade to 2021 returns 54,875 papers for 'climate change' and 284,055 for 'mental health', only 478 are returned for both terms[9].

This picture is rapidly changing. Research is racing to provide the evidence base required to understand, monitor and respond to the nature, prevalence and severity of climate-related impacts on mental health outcomes. A PubMed search for articles published on climate change and mental health in 2022 and 2023 returns 497 novel papers, and articles on climate anxiety abound in international media. The 2022–2023 Intergovernmental Panel of Climate Change (IPCC) reports included mental health impacts for the first time, with mentions throughout of co-beneficial actions for mental health, and climate adaptation and mitigation[10]. Recommendations for action by different stakeholders have also been outlined in recent academic and policy literature including by the World Health Organization[11].

While the climate and mental health research field is rapidly expanding, it remains disconnected, unequal and siloed across sectors, disciplines, and regions which hinders sharing, learning, and working cooperatively. A scoping review of 120 original studies published between 2001 and 2020 revealed that more than 77% of research that examined mental health in the context of climate-related exposures is conducted in high-income settings with a distinct lack of representation from diverse disciplines and affected communities[12].

To address these challenges,

Connecting Climate Minds was founded

Connecting Climate Minds is a global Welcome-funded initiative that aims to catalyse a connected, collaborative, and multidisciplinary climate change and mental health field. By fostering communities of practice across seven global regions, we conducted 18 virtual dialogues and 4 in-person dialogues (in Peru, Nigeria, India, and Cameroon) with over 600 experts in 80+ countries. This included people with lived experience and from communities with particular vulnerabilities and sources of resilience – small farmers and fisher peoples, Indigenous communities and young people. Through dialogues, surveys and expert consultations, we have developed aligned and inclusive agendas for research and action. These contain priority research themes to enable targeted evidence-based research and policy action for people who are experiencing the compounding mental health burdens of climate change.

How to define the terms climate change and mental health?

Climate change, mental health and their intersection are complex and wide-ranging fields. The terminology used to describe climate-related mental health problems varies around the world and across cultures, including Indigenous communities. Thus culturally appropriate use of terminology must be considered for both research and interventions. There is a clear need for guidance in understanding and defining climate-specific mental health experiences, what outcomes are relevant to measure, and what experiences are novel to the context of climate change. Additionally, researchers need to consider how others across disciplines and sectors may understand the terms they are using – for instance, 'resilience' might mean something very different to a climate expert or a mental health expert – by combining understandings new insights can emerge.

For Connecting Climate Minds mental health challenges were defined as thoughts, feelings, and behaviours that affect a person's ability to function in one or more areas of life and often involve significant levels of psychological distress. This includes but is not limited to, anxiety, depression, post-traumatic stress, psychosis, suicidal thoughts and substance abuse. In some contexts, stigma around mental health remains prevalent and may act as a barrier to accessing mental health and psychosocial support after climate disasters.

By experiences of the effects of climate change, we mean:

Experiencing direct impact of climate hazards, such as more frequent and intense heatwaves, wildfires, bushfires, drought, floods or storms (e.g., typhoons, hurricanes, cyclones).

Experiencing disruption to the social and environmental determinants of good mental health, such as being forced to move home, not being able to access food or water, losing livelihood or homelands, or disruption to cultural practices because of climate change.

Mental health challenges in the context of climate change:

How climate change may contribute to the prevalence or impact of existing mental health challenges.

How climate change may impact treatment access or effectiveness for those with mental health challenges.

How climate change may lead to worsening pre-existing mental health challenges.

How climate change may lead to new mental health challenges.

What are the **key insights** for research on climate and mental health?

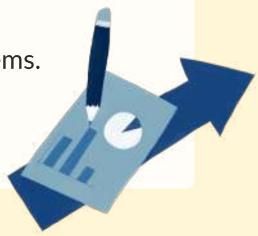
The impacts of climate change on mental health are widespread and deeply interconnected, with diverse pathways and mechanisms. In many regions, dialogue participants reported climate-related mental health experiences that have not yet been adequately captured in the evidence base for their region. Pathways of impact included direct experiences of climate hazards and their aftermath, such as social isolation in extreme heat, food, and water insecurity, the rising cost of living, domestic and sexual violence, increased risks of conflicts, forced migration from lands, and loss of cultures. Further, participants collectively shared their concerns about experiences of compounding climate hazards and traumas, as they have less time to recover and heal from one climate-related crisis before the next one hits.

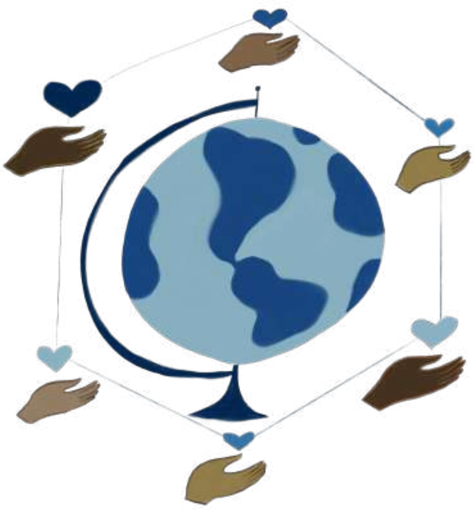
Examples of emerging impacts and mechanisms inadequately captured in existing literature:

Sea level rise and coastal erosion compound impacts on the mental health of coastal communities and fisherpeople, who are an underappreciated and vulnerable group.

Women farmers experience increased domestic violence as they are inaccurately held accountable for worsening crop yields caused by increasingly unpredictable climates and prolonged drought.

Extreme heat leads to social isolation as people can't leave their homes to see family and friends or exercise, worsening their mental health. This includes people with pre-existing mental health problems. Psychiatrists in the dialogues reported the impact of heat-related social isolation on the outcomes of their patients.





Many of the most climate-vulnerable settings have the least access to mental healthcare. The level of community cohesion and social support was a key reported differentiator between people who were affected by a climate hazard and remained able to cope versus those who were severely psychologically and functionally affected. There was a great value and demand for community-led mental health support to meet the needs of the most affected communities, to ensure sustained support for people with mental health challenges after climate disasters, and to build psychosocial resilience in communities as a preventative measure and pillar of climate adaptation. Intergenerational community gatherings were also described as a co-beneficial protective factor for health, culture, and environmental actions.

Examples of research priorities emerging from Connecting Climate Minds

Identifying and understanding the mental health impacts of climate change and what factors may increase risk to or be protective against these impacts

For example, identifying which mental health challenges are increasing in frequency or severity as a result of climate change, or understanding at what point climate-related anxiety and stress become debilitating and impact functioning such that they constitute a mental health challenge.

Mental health solutions in the context of climate change

For example, understanding and evaluating post-disaster support to protect against and respond to mental health impacts of experiencing extreme weather and climate events, or integrating Indigenous ways of knowing, being and doing into conceptual frameworks for understanding and responding to mental health impacts of climate change.

Mental health benefits and trade-offs of climate action

For example, understanding mechanisms of multi-sectoral collaboration and knowledge sharing to integrate mental health into climate actions, or identifying and evaluating the impact of climate mitigation and adaptation actions on mental health.

Pathways and mechanisms through which climate change impacts mental health

For example, assessing the impact of heat on the bio-availability of medicines used to treat mental health, or understanding how climate change disrupts key educational events in the lives of young people, and how this contributes to their vulnerability to mental health issues.

What are the key emerging insights for action on climate and mental health?

Many relevant insights and interventions for the mental health and psychosocial impacts of the climate crisis already exist in communities and other relevant disciplines, but there is an urgent need for investment in identifying, evaluating, amending, scaling up and rolling out relevant interventions. Awareness raising and education of healthcare professionals, non-profit organisations and decision-makers across both climate and mental health are further required to improve decision-making that can benefit climate change and mental health together. In particular, education and communication about climate change must consider the mental health implications of raising awareness of climate threats and integrate appropriate support.

Research and action on climate change and mental health must be led by or with affected communities, with an appreciation for sovereignty of knowledge and data by traditional owners of lands, including Indigenous Peoples. Further, there is a need for standardisation of metrics and methods to measure the relevant climate-related exposures and mental health outcomes in assessing the impacts of climate change on mental health and co-benefits of climate adaptation and mitigation actions and climate-relevant mental health support. This standardisation needs to be balanced with local contextualisation to honour diverse understandings and definitions of climate change and mental health.

Finally, there is a strong desire for continued investment in the communities of practice within Connecting Climate Minds. Not only would funding enable capacity building for the range of stakeholders identified by the dialogues across disciplines, sectors and nations, but it would also foster further engagement in enacting the research and action agenda to achieve better outcomes for mental health and the climate.

Connecting to further resources

The collective vision of Connecting Climate Minds is to facilitate shared solutions that enable people and the planet to thrive, through the power of connection. This includes appreciating the interconnections between climate change and mental health, bringing together diverse disciplines and sectors globally and connecting researchers with local decision-makers. Thus, people working in relevant research and practice become aware of and account for the implications of the climate–mental health nexus in their work. Together, we can protect the mental health of more people globally from the impacts of the climate crisis and build psychological resilience for sustained climate action.

You can join us by:

Deepening your understanding of the interlinkages between climate change and mental health by checking out all the published research and policy briefing papers developed by the Climate Cares Centre, the World Health Organisation and the American Psychological Association (APA) 2022 and 2021 reports. Further, mental health is discussed in key climate and health reports including the Lancet Countdown reports, the World Economic Forum Health Impacts report, and the IPCC Sixth Assessment Report adaptation, mitigation and synthesis reports.

Signing up to the Global Online Hub to connect with others interested in the climate and mental health field from around the world, share learnings, and form partnerships to enact the research and action agendas on climate and mental health.

Engaging with and amplifying the outputs of Connecting Climate Minds, including the research and action agendas, lived experience stories, case studies and toolkits available on the Global Online Hub. Use the research and action agendas to inform your work in this space, and share with your networks.

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